

The effect of a herbal paste and oil extract on the lipid content of canine hair fibres

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Background – Application of herbal paste and oil to a dog's coat and body before rinsing (often combining with shampooing) is a cosmetic therapy available in Japan. It is highly appreciated by users, who claim that the treatment makes the coat shinier, improves volume and eliminates tangles. However, there has been no scientific evaluation of such treatments.

Hypothesis/Objectives – Improvement of hair condition is derived from oils such as sebum and conditioning oils because chemicals are not used. Therefore, we examined nonpolar lipids (the primary lipids in dog hair) and the botanical oils used in this therapy.

Animals – Hair samples were obtained from six beagle dogs.

Methods – Groups were based on different combinations of the following processes: rinsing, shampooing, herbal therapy and herbal therapy with oil extract. Analysis of lipids was performed by high performance thin layer chromatography.

Results – The processes of shampooing and herbal therapy were associated with an equivalent reduction in cholesterol ester and triglyceride (TG). However, hair treated by herbal therapy combined with oil extract had an almost three-fold higher TG content, even after shampooing.

Conclusions and clinical importance – This study demonstrated that the herbal therapy was able to coat hair samples with TG that was not removed with rinsing. Further investigation is required to evaluate the possible benefits of the application of botanical products containing lipids, such as TG, on hair coat quality in dogs.

Introduction

The Sanskrit term “ayurveda” means life knowledge and “ayurvedic medicine” is a medical system with its roots in the Indian subcontinent. Some ayurvedic therapies and practices contributing to wellness have found medical uses.¹ Herbal compounds are an essential part of these therapies. In humans, application of herbal paste to the face and body is performed as ayurvedic beauty therapy. In a similar way, the application of herbal paste has been developed for animals based on ayurvedic medicine. The therapy is known as the Animal Aryurveda® Herb pack (Eureka Co Ltd.; Kanagawa, Japan) and is commercially available (Figure 1a).

Cosmetic properties such as texture, volume and gloss are mostly related to the hair cuticle, which is the surface layer or outer barrier of hair fibres. The texture of hair depends on the smoothness of the hair surface. Hair volume is partly dependent on the amount of tangles. In

other words, the volume decreases as the hair coat contains more knots and tangles. Both hair texture and volume are influenced by friction at the hair surface. In physical terms, hair gloss is an optical property and is also related to the surface properties of hair. Gloss is dependent on reflection of light from hair fibres; it is generally assumed that there is no satisfactory way to evaluate hair gloss.² We presumed that improvement in cosmetic features of canine hair might be caused by agents which can maintain or coat the hair surface, such as hair conditioners. The Animal Aryurveda® Herb pack contains two herbal powders and oil extract (Figure 1b); we hypothesized that its potential cosmetic effects might be derived from lipids which naturally have a conditioning effect on the hair surface. It should be noted that this product does not contain any detergents or industrial lubricants, such as simethicone.

The lipids in human hair are reported to include primarily free fatty acids, triglyceride, phytosphingosine, ceramide, cholesterol oleate, free cholesterol and cholesterol sulfate.³ In a study of the lipids in dog hair, it was concluded that the same lipids might be derived primarily from the skin surface, given that dog hair lipids include sterol esters, wax diesters, free sterols, polar lipids and unidentified components.⁴

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Figure 1. Aryurveda herbal therapy. a) The dog is washed and herbal paste is applied for 5–10 minutes then rinsed off with a shampoo. b) The herbal paste comprises herbal powder and oil mixed with water.

Lipids play an important role in determining the functional and structural properties of hair. In cosmetology, nonpolar lipids such as cholesterol (CHOL), cholesterol esters (CE), wax esters (WE), triglyceride (TG) and free fatty acids (FFA) have attracted attention with regard to maintaining hair health, because detergents in shampoo products remove free lipids including nonpolar lipids and ceramides. In particular, triglyceride-containing lipids are commonly believed to contribute to hair gloss and lubrication.⁵

Among professional pet groomers in Japan, the Animal Aryurveda® Herb pack is reputed to improve the texture, volume and gloss of the coat. The components responsible for these effects have not been investigated. In the present study, we used the components of the pack to assess the impact on nonpolar lipids in hair.

Materials and methods

Hair treatment procedures

Hair samples were obtained from six pet, healthy, clinically normal, female neutered beagles that ranged in age from 5 to 9 years (mean: 7 years 11 months) and belonged to colleagues of the authors. None of the dogs had any history or signs of disease. The dogs were not bathed for two weeks and then a hair sample was carefully obtained by trimming at the skin surface with scissors. The hair samples were stored in a freezer at -20°C until assay, which was performed at room temperature ($23 \pm 1^{\circ}\text{C}$) and a relative humidity of $55 \pm 6\%$.

The herbal formulation (provided by Eureka Co. Ltd.) contained herbal powder and oil extract. The herbal powder was made from the leaves of Neem (*Azadirachta indica*) and *Cassia auriculata*, whereas the oil extract was extracted from the same herbs in a sesame oil base. The study conditions were designed to simulate use of the Aryurveda Herb pack for grooming by testing canine hair strands *in vitro*. Herbal therapy consists of four steps, which are application of the herbal paste, shampooing, rinsing and drying. Based on these steps, the hair samples were treated as follows. According to the instructions of the manufacturer, 30 g of herbal powder was placed in a plastic bottle containing 200 mL of warm tap water (40°C) and shaken vigorously (Figure 1b). In some experiments the paste was mixed with 3 mL of oil extract and left to stand for 5 min. Then a hair sample (200 mg) was rubbed gently with the paste in the same manner as if this therapy was being performed on an animal; the hair being soaked in

a beaker for 10 min. To simulate the shampooing process, the hair sample was then soaked in 10% sodium lauryl sulfate (SLS) and stirred continuously for 10 min using a magnetic stirrer. Before shampooing hair treated with herbal paste, the paste was removed from the hair sample by briefly placing it under running tap water. In this step, a tea strainer was used to avoid loss of hair. To simulate the rinsing process, the hair sample was soaked in tap water, which was changed after stirring for 2.5 min. This process was performed twice. For drying, the drained hair sample was transferred to filter paper and was dried with a hair dryer for 10 min.

In order to investigate factors that influenced the lipid content of the hair coat, we tested the effects of different combinations of processes:

First group: four processes performed: application of the herbal paste with both powder and oil, shampooing, rinsing and drying (herb-oil-shampoo group)

Second group: three processes: application of the herbal paste with both powder and oil, rinsing and drying (herb-oil group)

Third group: four processes: application of the herbal paste powder only, shampooing, rinsing and drying (herb-shampoo group)

Fourth group: application of the herbal paste powder only followed by rinsing and drying (herb group)

Fifth group: shampooing, rinsing and drying (no application of herbal paste) (shampoo group)

Sixth group only received rinsing and drying (rinse group: control).

Additional analysis was performed to clarify the contribution of the oil extract to herbal therapy. The herb-oil-shampoo group was compared with an additional group (oil-shampoo group), in which the hair coat was treated only with the oil extract. In the oil-shampoo group, a hair sample (150 mg) was dipped in 500 μL of oil extract for 10 min, followed by the same procedures as in the herb-shampoo group. The shampooing process was modified by reducing the SLS concentration to 2.5% and the frequency of shampooing was set at 0 times (skipping the process), once or twice, with shampooing duration 5 min per time. When the hair sample was shampooed twice, it was rinsed under running tap water before the second shampooing. In the herb-oil-shampoo group, the modified shampooing process, and the rinsing and drying processes were carried out sequentially, as described above.

Extraction of lipids

In the oil-shampoo group, hair samples that had not undergone the shampooing process were pressed between two pieces of filter paper for 5 min using a vice to absorb excess oil from the hair. These samples from the latter group and all dried hair samples from the other groups, were treated as follows: 100 mg of dried hair was placed in 4 mL of chloroform/methanol (2:1 by volume) and shaken vigorously for 2 h at 37°C ; after the hair was removed, the extract was dried under a stream of nitrogen at 38°C and then reconstituted in 60 μL of chloroform/methanol (2:1) for analysis. An aliquot (10 μL) of the reconstituted lipid extract was pipetted onto a high performance thin layer chromatography (HPTLC) plate (Merck; Tokyo, Japan). In order to analyse the oil extract, 0.5 mL oil extract diluted in chloroform/methanol was subjected to HPTLC. According to the previously described method, the chromatogram was first developed to the 9 cm level with hexane, then to the 9 cm level with benzene and finally to the 7 cm level with hexane/benzene/acetic acid (50:50:1).⁶

For development of colour, the HPTLC plates were air-dried, sprayed with 10% CuSO_4 followed by an 8% aqueous solution of phosphoric acid and heated at 180°C on a hot plate for 10 min. Images of the plates were captured with a LAS 4000 Image Reader (FujiFilm; Tokyo, Japan) and density plot analysis was performed using the Multi Gauge v3.2 programme (FujiFilm). Bands in the hair extracts corresponding to CHOL, CE, TG and FFA were defined by comparison with lipid standards, including cholesterol, cholesteryl oleate for CE, triolein for TG and palmitic acid for FFA (Sigma-Aldrich; Tokyo, Japan).

Statistical analysis

In order to evaluate relative changes of each lipid, the band densities in the control group (rinse group) were set as 100%, and densities in the other groups were compared with the control group. The results (%) are reported as the mean $\pm$ SD. Normality test was performed using the D'Agostino–Pearson test. Statistical analyses were conducted by using a one-way factorial ANOVA or a one-way repeated measures ANOVA, and verified by Tukey's multiple comparison test. Spearman's rank correlation test was used for correlation analyses between shampooing frequencies and TG contents in the shampooing processes. The level of significance was set at $P < 0.05$. All analyses were performed by using GraphPad Prism[®] 5.2 software (GraphPad Inc.; San Diego, CA, USA) and SPSS v18 (SPSS, IBM; Somers, NY, USA).

Results

Factors contributing to changes in lipid content of the hair samples

HPTLC was performed to evaluate the factors contributing to improvement of the hair coat by herbal therapy (Figure 2). After herbal therapy, CHOL and FFA showed no significant changes compared with the control group (rinse group; Figure 3).

There were significant changes for CE (one-way factorial ANOVA, $P < 0.05$). In detail, there was a significant decrease of CE in the herb-oil group ($48\% \pm 18\%$, $P = 0.019$) and herb-oil-shampoo group ($43\% \pm 14\%$, $P = 0.008$). The CE was not reduced by either the application of herbal paste (without oil) and/or shampooing (shampoo $P = 0.564$; herb, $P = 0.749$; herb-shampoo, $P = 0.134$). Adding oil extract (herb-oil and herb-oil-shampoo groups) significantly decreased the CE content ($P = 0.019$ and $P = 0.008$, respectively).

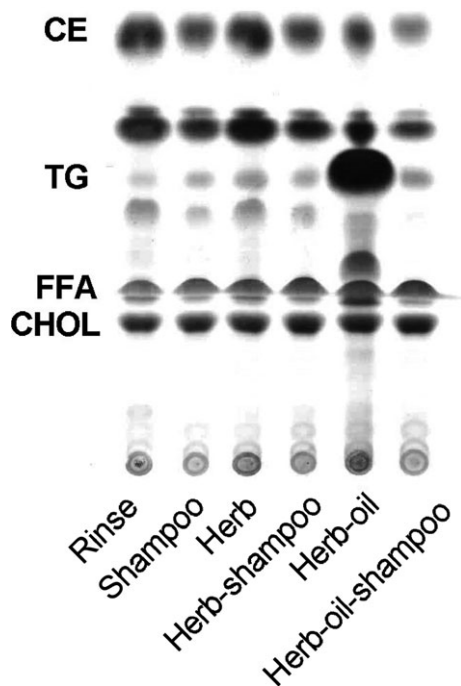


Figure 2. High performance thin layer chromatography image of the lipid content of hair samples following various applications of the Animal Aryurveda[®] herb pack, six comparative groups. Abbreviations: CE cholesterol ester, CHOL cholesterol, FFA free fatty acids, TG triglyceride.

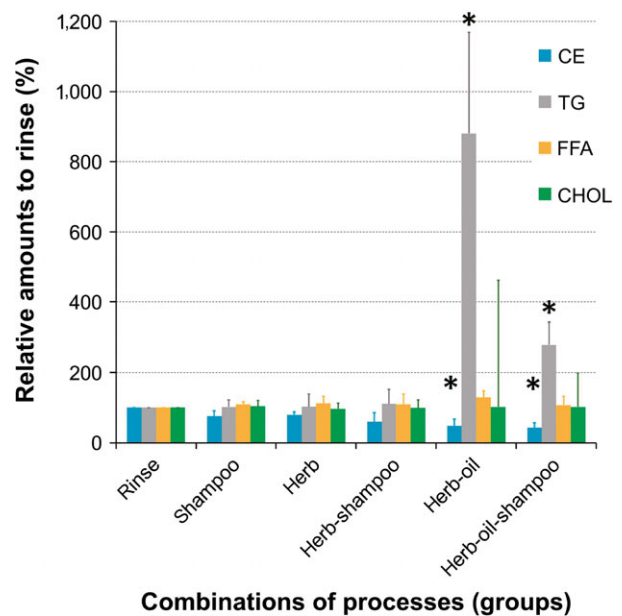


Figure 3. Comparison of fatty acid content in treated hair samples, with the control group. The difference in the content of each fatty acid from the control group (rinse only) expressed as a percentage by taking the value in the control group as 100%, using a density plot of images of high performance thin layer chromatography plates. Asterisks indicate significant differences comparing five groups with the rinse group ($P < 0.05$, one-way factorial ANOVA with Tukey's multiple comparison test). Abbreviations: CE cholesterol ester, CHOL cholesterol, FFA free fatty acids, TG triglyceride.

Triglyceride was markedly increased in the herb-oil group and the herb-oil-shampoo group; with significant increases compared with the control ($P < 0.0001$, $P = 0.003$, respectively), with an 880% and a 279% increase relative to the control, respectively (Figure 3). No statistically significant differences were observed for other groups ($P = 1.0$ for all). Adding oil extract appeared to influence the hair even after shampooing with SLS.

HPTLC analysis of the oil extract demonstrated that the TG fraction was the largest component of the extract and other lipids were minor components (Figure 4).

Contribution of oil extract to the hair coat

In order to evaluate the influence of TG in the oil extract, the effect of shampooing frequency was examined by comparing the herb-oil-shampoo group with the oil-shampoo group. In the herb-oil-shampoo group (Figure 5a) the TG content showed a markedly significant increase ($791 \pm 458\%$, $P = 0.006$) relative to the control group before shampooing, whereas it was significantly decreased after shampooing once ($223 \pm 109\%$, $P = 0.007$) compared with no shampooing (one-way repeated measures ANOVA, $P < 0.05$). Shampooing twice reduced the TG content to almost the same level as in the control group ($122 \pm 56\%$, $P = 0.457$). The TG in hair samples successively decreased after every shampoo treatment, and significant decreases were seen between no shampooing and shampooing once ($P = 0.021$), and between shampooing once and shampooing twice ($P = 0.005$).

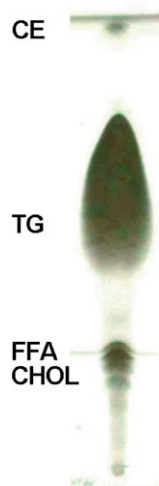


Figure 4. High performance thin layer chromatography of the oil extract from the Animal Aryurveda® Herb pack. Abbreviations: CE cholesterol ester, CHOL cholesterol, FFA free fatty acids, TG triglyceride.

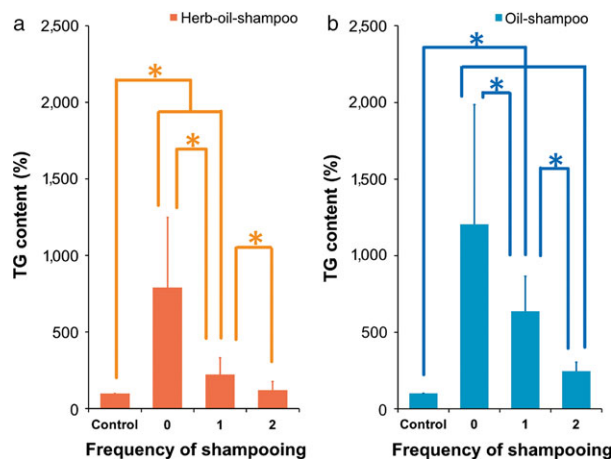


Figure 5. Comparison of triglyceride (TG) content between the herb-oil shampoo group (a) and oil-shampoo group (b). On the horizontal axis, Control is the rinse group and the numbers 0, 1 and 2 show the frequency of shampooing. Differences in the triglyceride content are expressed as percentages by taking the value in the rinse group (Control) as 100%. Asterisks indicate significant differences compared with the rinse group ($P < 0.05$, one-way repeated measures ANOVA with Tukey's multiple comparison test), $n = 6$.

In the oil-shampoo group (Figure 5b), the TG content also showed a marked increase ($1204 \pm 782\%$, $P < 0.0001$) before shampooing relative to the control. Reductions of TG were significant after shampooing once ($636 \pm 228\%$, $P = 0.003$) and twice ($247 \pm 57\%$, $P = 0.022$) compared with the control. Each repetitive shampoo treatment significantly decreased the TG contents from the prior one (0 to 1 time, $P = 0.001$; 1 to 2 time, $P = 0.006$). There was an inverse correlation between the TG content and the frequency of shampooing by Spearman's rank correlation analysis ($P < 0.05$). However, a significant proportion of the TG fraction was still detected after shampooing twice ($247 \pm 57\%$, $P = 0.022$).

Discussion

Historically, botanical products have been used in cosmetic therapy and their efficacy has usually been assessed empirically. In the present study, we examined the botanical therapy Animal Aryurveda® Herb pack, in relation to its effects on the lipid component of the hair coat. Herbal products can have a cleansing effect on the lipid content of dog hair because their activity may resemble 10% SLS detergent. Moreover, the herbal paste plus extract oil contributed TG to the hair coat and this may be a beneficial cosmetic effect of this product.

We assumed that application of herbal paste therapy, followed by shampooing, would prevent removal of hair lipids, because the paste is not a detergent, but would still remove dirt and excess lipid from the hair surface. However, we found that application of the herbal paste was associated with removal of cholesterol, cholesterol esters and free fatty acids as effectively as using 10% SLS. Dramatic results were seen in two groups treated with oil extract plus herbal paste with a marked increase of TG in the hair fibres. HPTLC analysis (Figures 3 and 4) suggested that elevation of TG in the treated hair fibres may have been due to TG from the oil extract. The other fatty acids showed no significant increase with herbal therapy, and the CE content decreased significantly in the herb-oil group and herb-oil-shampoo group. The combination of herbal paste and oil might effectively remove fatty acids from the hair surface, and the CE fraction was small in the oil extract whereas other fatty acids existed in larger amounts (Figure 4). Part of the CE on the hair surface is derived from sebum and might be removed by application of the herbal pack with oil extract. Based on the present results, we suggest that any presumed cosmetic benefits may in part be due to supplying TG to the hair coat.

We evaluated the TG content of the hair samples after treatment, including the oil extract. In order to gently remove the oil extract, the SLS concentration was set so as not to damage the hair cuticle. Both 1% and 2.5% SLS solutions have been used; the latter concentration was chosen in this study.^{7,8} In the oil-shampoo group, hair samples were treated directly with the oil extract to evaluate its effect. After treatment, the excess oil between hair fibres and on the hair surface was removed by pressing the hair samples between two pieces of filter paper, which resulted in varying quantities of oil extract remaining in the hair samples. Consequently, the standard deviation in the TG content was larger in the oil-shampoo group after no shampooing and shampooing once (no shampoo, $SD=782\%$; once, $SD=228\%$) than in the herb-oil-shampoo group (no shampoo, $SD=458\%$; once, $SD=109\%$). In both groups, the decrease of the TG content was correlated with the frequency of shampooing, suggesting that TG in the hair samples originated from the oil extract. Thus, herbal therapy with oil extract may add TG to the hair surface.

Oil extracts from *Camellia japonica* (camellia oil), coconut oil, red palm oil (extracted from *Elaeis guineensis*), avocado oil and other oils are believed to penetrate into human hair strands and are used as ingredients of hair treatments to improve hair gloss. The fatty acid TG is

found in botanical oils and a hair conditioning function has been suggested; for example, it has been reported to inhibit water-induced swelling of damaged hair^{9,10,11} and is used as a conditioning agent for protection and repair of damaged hair. Treating canine hair with an oil extract may improve hair coat lubrication along with an increase of glossiness; it may also prevent tangling in animals with curly coats such as poodles. Removing tangles from a curly coat generally leads to an increase of hair coat volume and less damage to the hair cuticles. However, initially, further studies are required to validate methods for assessing canine hair coat gloss, texture and volume. Subsequently, the investigation of the potential benefits of herbal products, rich in TG, on canine hair coat quality might then be feasible and include assessment of quantitative effects.

References

1. US Department of Health and Human Services, National Institutes of Health. Ayurvedic Medicine: In Depth. Available at: <https://nccih.nih.gov/health/ayurveda/introduction.htm>. Accessed February 24, 2017.
2. Czeplch W, Hohm G, Tolkiehn K. Gloss of hair surfaces: problems of visual evaluation and possibilities for goniophotometric measurements of treated strands. *J Soc Cosmet Chem* 1993; 44: 299–318.
3. Lee W. Integral hair lipid in human hair follicle. *J Dermatol Sci* 2011; 64: 153–158.
4. Sharaf DM, Clark SJ, Downing DT. Skin surface lipids of the dog. *Lipids* 1997; 12: 786–790.
5. Fregonesi A, Scanavez C, Santos L et al. Brazilian oils and butters: the effect of different fatty acid chain composition on human hair physiochemical properties. *J Cosmet Sci* 1999; 60: 273–280.
6. Kirby NA, Hester SL, Rees CA et al. Skin surface lipids and skin and hair coat condition in dogs fed increased total fat diets containing polyunsaturated fatty acids. *J Anim Physiol Anim Nutr (Berl)* 2009; 93: 505–511.
7. Rele A, Mohile RB. Effect of mineral oil, sunflower oil, and coconut oil on prevention of hair damage. *J Cosmet Sci* 2003; 54: 175–192.
8. Ruetsch SB, Kamath YK, Rele AS et al. Secondary ion mass spectrometric investigation of penetration of coconut and mineral oils into human hair fibers: relevance to hair damage. *J Cosmet Sci* 2001; 52: 169–181.
9. Swift JA, Smith JR. Atomic force microscopy of human hair. *Scanning* 2000; 22: 310–318.
10. Bellare J, Iyer R, Mainkary AR et al. A study on the conditioning effects of natural shampoos using the scanning electron microscope. *Int J Cosmet Sci* 2001; 23: 139–145.
11. Kojima T, Tsuji S, Niwa M. Distribution analysis of triglyceride having repair effect on damaged human hair by TOF-SIMS. *Int J Polym Anal Ch* 2012; 17: 21–28.

Résumé

Contexte – L'application de pommade et d'huiles végétales sur le pelage et la peau d'un chien avant rinçage (souvent en association avec un shampoing) est un traitement cosmétique disponible au Japon. Il est hautement apprécié par les utilisateurs qui rapportent que le traitement rend le pelage brillant et améliore le volume et le démêlage. Cependant, aucune évaluation scientifique n'existe à ce sujet.

Hypothèses/Objectifs – L'amélioration du pelage est dérivé d'huiles tels que le sébum et les huiles après-shampoing, les produits chimiques n'étant pas utilisés. Ainsi, nous examinons les lipides non polaires (les lipides primaires du poil de chien) et les huiles végétales utilisés dans ce traitement.

Sujets – Les échantillons de poil ont été prélevés sur six chiens beagle.

Méthodes – Les groupes ont été formés par différentes combinaisons des procédures suivantes : rinçage, shampoing, traitement aux herbes et traitement aux herbes avec extrait d'huile. L'analyse de lipides a été réalisée par chromatographie à haute performance sur couche mince.

Résultats – Les procédures de shampoing et de traitement aux herbes étaient associées avec une diminution équivalente d'ester de cholestérol et de triglycéride (TG). Cependant, les poils traités par les herbes combinées à l'extrait d'huile avait presque trois fois plus de TG même après shampoing.

Conclusions et importance clinique – Cette étude démontre que le traitement végétal est capable d'enrober les échantillons de poil avec des TG qui n'était pas enlevé par rinçage. D'autres études sont nécessaires pour évaluer les bénéfices possibles de l'application de produits botaniques contenant des lipides, tels que les TG, sur la qualité du pelage des chiens.

Resumen

Introducción – La aplicación de pasta y aceite de hierbas al pelo y cuerpo de un perro y el cuerpo antes de enjuagar (a menudo en combinación con la aplicación de champú) es una terapia cosmética disponible en Japón. Es muy apreciado por los usuarios, que afirman que el tratamiento hace que el pelaje sea más brillante, mejora el volumen y elimina enredos de los pelos. Sin embargo, no ha habido evaluación científica de tales tratamientos.

Hipótesis/Objetivos – La mejora de la condición del cabello deriva de aceites como el sebo y los aceites acondicionadores porque no se utilizan productos químicos. Por lo tanto, se examinaron los lípidos no polares (los lípidos primarios en el pelo del perro) y los aceites botánicos utilizados en esta terapia.

Animales – Se obtuvieron muestras de pelo de seis perros de raza Beagle.

Métodos – Los grupos se basaron en diferentes combinaciones de los siguientes procesos: enjuague, lavado con champú, terapia con hierbas y terapia con extracto de aceite de hierbas. El análisis de lípidos se realizó mediante cromatografía de capa fina de alta resolución.

Resultados – Los procesos de lavado con champú y terapia de hierbas se asociaron con una reducción equivalente en éster de colesterol y triglicéridos (TG). Sin embargo, el cabello tratado con hierbas combinado con extracto de aceite tenía un contenido de TG casi tres veces mayor, incluso después de la aplicación de champú.

Conclusiones e importancia clínica – Este estudio demostró que la terapia con extractos de hierbas fue capaz de cubrir las muestras de pelo con TG que no se eliminaron con el enjuague. Se requiere una investigación adicional para evaluar los posibles beneficios de la aplicación de productos botánicos que contienen lípidos, como TG, sobre la calidad del pelo en perros.

Zusammenfassung

Hintergrund – In Japan gibt es eine kosmetische Therapie, bei der eine pflanzliche Paste und Öl auf das Fell und Körper des Hundes aufgetragen werden, bevor diese abgewaschen (oft in Kombination mit Shampoo) wird. Die Anwender sind davon begeistert und betonen, dass das Haarkleid dadurch glänzender wird, sich an Volumen verbessert und Haarverfärbungen weniger häufig auftreten. Es gibt jedoch bisher keine wissenschaftliche Evaluierung derartiger Behandlungen.

Hypothese/Ziele – Die Verbesserung des Haarkleids wird auf Öle wie Sebum und Konditionierer zurückgeführt, da keine Chemikalien im Einsatz sind. Daher haben wir nicht-polare Lipide (die Primärlipide im Hundehaar) und die botanischen Öle, die bei dieser Behandlung eingesetzt werden, untersucht.

Tiere – Es wurden von sechs Beagles Haarproben genommen.

Methoden – Die Gruppen wurden der Therapie entsprechend eingeteilt: abspülen, shampooen, pflanzliche Therapie und pflanzliche Therapie mit Ölextrakten. Eine Analyse der Lipide wurde mittels High Performance Dünnschichtchromatografie durchgeführt.

Ergebnisse – Das Procedere des Shampooens und der pflanzlichen Therapie wurde mit einer entsprechenden Reduzierung der Cholesterolester und Triglyzeride (TG) im Zusammenhang gesehen. Haare, die mit pflanzlicher Therapie mit Ölextrakten behandelt worden waren, zeigten jedoch, selbst nach dem Shampooen, einen fast dreifach höheren TG Gehalt.

Schlussfolgerungen und klinische Bedeutung – Diese Studie zeigte, dass die pflanzliche Therapie imstande war, die Haarproben mit TG einzuhüllen, welche durch das Abspülen nicht wieder entfernt wurden. Eine weitere Untersuchung ist nötig, um die möglichen Vorteile der Applikation von pflanzlichen Produkten, welche Lipide, wie TG, enthalten, auf die Haarqualität von Hunden, zu untersuchen.

要約

背景 – 日本では、リンス前にハーブペーストとオイルを犬の被毛や体へ塗布する美容療法(シャンプーと組み合わせることが多い)が利用可能である。この療法は、被毛の光沢や量が増し、毛のもつれを改善されるとして、オーナーから高く評価されている。しかしながら、このような治療法について科学的検証はなされていない。

仮説/目的 – 本治療には、化学物質が使用されないため、被毛状態の改善は、皮脂およびコンディショニングオイルなどの油に由来している。したがって我々は、非極性脂質(犬の被毛の主要脂質)および本治療で使用された植物性油を調査した。

供与動物 – 6頭のビーグル犬から得た被毛サンプル。

方法 – リンス、シャンプー、ハーブ療法、オイル抽出液を併用したハーブ療法の組み合わせにより各グループを作成した。脂質の分析は、高性能薄層クロマトグラフィーを用いて実施した。

結果 – シャンプーおよびハーブ療法では、コレステロールエステルおよびトリグリセリド(TG)が同程度に減少した。しかし、オイル抽出液を使用したハーブ療法では、被毛のTG含有量がほぼ3倍で、これはシャンプー後であっても同じであった。

結論および臨床的な重要性 – 本研究は、ハーブ療法が、リンスで除去されなかったTGを被毛サンプルにコーティングすることができたことを証明した。TGなどの脂質を含む植物性製品の塗布が、犬の毛並みもたらす潜在的な有益性を評価するために、さらなる調査が必要である。

摘要

背景 – 对犬的皮毛和身体涂抹草药膏和油(经常结合香波洗澡),然后冲洗,是日本的一种的美容治疗。这受到使用者的高度赞扬,他们表示治疗使得毛发更光亮,增加了毛发总量,并去除了打结的毛。但是,这种治疗并未得到科学性的评估。

目的 – 皮脂为毛发提供油脂,可改善毛发状态,由于调理油是化学物质而未使用。因此,我们检测在治疗中使用的非极性脂质(犬毛发中的重要脂质)和植物油。

动物 – 从六只比格犬处获得毛发样本。

方法 – 基于如下过程而分组:冲洗、香波洗浴、草药治疗、草药与萃取油治疗。使用高分层薄层色谱分析法分析脂质。

结果 – 香波洗浴和草药治疗过程,使毛发胆固醇和甘油三酯等量减少。但是,经过草药与萃取油治疗的毛发中,甚至是在用香波洗浴后,甘油三酯含量几乎高了三倍。

结论和临床意义 – 研究表明,草药治疗能够使毛发中的甘油三酯不随着冲洗而流失。用含甘油三酯这类脂质的植物产品涂抹,对改善犬被毛质量带来的好处,需要更多调查评估。

Resumo

Contexto – A aplicação de pasta herbal e óleo na pelagem e no corpo de cães antes do enxague (muitas vezes associado ao xampu) é um tratamento cosmético disponível no Japão. Esta terapia é muito apreciada pelos usuários, que relatam o maior brilho da pelagem, melhora no volume e eliminação de nós. Entretanto, a avaliação científica destes tratamentos não foi realizada.

Hipótese/Objetivos – A melhora na condição da pelagem é derivada dos óleos tipo sebo e óleos condicionadores, já que substâncias químicas não são utilizadas. Desta forma, lipídeos apolares (os principais lipídeos na pelagem dos cães) e os óleos botânicos usados nesse tratamento foram analisados.

Animais – Amostras de pelos foram obtidas de seis cães da raça beagle.

Métodos – Os grupos foram separados baseado em diferentes combinações dos seguintes processos: enxague, xampu, terapia herbal e terapia herbal com extrato de óleo. A análise dos lipídeos foi realizada por cromatografia em camada fina de alta performance.

Resultados – Os processos de lavagem com xampu e terapia herbal foram associadas a uma redução equivalente de colesterol e triglicerídeos (TG). Entretanto, o pelo tratado com a terapia herbal combinada com o extrato de óleo obteve uma cobertura de TG significativamente maior, mesmo após lavagem com xampu.

Conclusões e importância clínica – Este estudo demonstrou que a terapia herbal foi capaz de fazer uma cobertura de TG nas amostras de pelo, que não foi removida após enxague. Estudos posteriores são necessários para avaliar os possíveis benefícios da aplicação de produtos botânicos contendo lipídeos, com TG, na qualidade da pelagem de cães.